

LOGICAL STRUCTURES

Review session for Midterm test

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Natural Deduction

Question 1. Write out a proof term for the corresponding implication or indicate that it is not derivable

- ① $A \rightarrow B \vee C \dashv\vdash (A \rightarrow B) \vee (A \rightarrow C)$
- ② $(A \rightarrow B) \rightarrow C \dashv\vdash (A \vee C) \wedge (B \rightarrow C)$

Normal Form

Question 2. Convert the following formulas into Conjunctive Normal Form using and without using CNF procedure. In each case, determine its validity.

① $\phi = (p \rightarrow (q \rightarrow r)) \rightarrow (p \rightarrow (r \rightarrow q))$

② $\psi = \neg(p \rightarrow q) \vee ((r \vee s) \rightarrow (q \vee t)) \vee (\neg p \rightarrow \neg v)$

Question 3. Apply algorithm HORN to the following Horn formula

- ① $(p \wedge q \wedge w \rightarrow \perp) \wedge (t \rightarrow \perp) \wedge (r \rightarrow p) \wedge (\top \rightarrow r) \wedge (\top \rightarrow q) \wedge (u \rightarrow s) \wedge (\top \rightarrow u)$
- ② $(\top \rightarrow q) \wedge (\top \rightarrow s) \wedge (w \rightarrow \perp) \wedge (p \wedge q \wedge s \rightarrow \perp) \wedge (v \rightarrow s) \wedge (\top \rightarrow r) \wedge (r \rightarrow p)$

Question 4. For each formula ϕ of the following formulas, draw DAG of $T(\phi)$.

- ① $\phi = (p \rightarrow q) \vee (r \rightarrow p)$. Prove ϕ is valid.
- ② $\phi = ((p \vee (q \vee r)) \wedge ((p \vee \neg q) \wedge ((q \vee \neg r) \wedge ((r \vee \neg p) \wedge (\neg p \vee (\neg q \vee \neg r))))))$.
Prove ϕ is not satisfiable.